

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
DATA SCIENCE AND ANALYTICS
(7th SEMESTER, 4TH Year) LAB EXPERIMENT # 3

Lab # 3: Introduction to Data Manipulation Using Pandas

Objective: The objective of this lab session is to introduce students to the fundamental concepts Pandas libraries in Python. Students will learn how to create and manipulate and use Pandas for data handling, analysis, and visualization of structured data.

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. **Pandas:** Create and work with Pandas Series and DataFrames, perform basic data analysis, handle missing values, and read/write data using CSV and Excel files.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Pandas: Dataset Manipulation

In Data Science and Analytics, data is collected from various sources such as surveys, sensors, websites, databases, spreadsheets, and IoT devices. However, raw data is often incomplete, inconsistent, duplicated, or stored in different file formats. Before meaningful analysis or machine learning can be performed, this data must be cleaned, organized, and transformed.

Pandas is a powerful open-source Python library designed to simplify these tasks. It provides highperformance data structures and a rich collection of functions for importing, cleaning, manipulating, analyzing, and exporting structured data. Because of its flexibility and ease of use, Pandas has become one of the most widely used libraries in Data Science, Machine Learning, Artificial Intelligence, Finance, Healthcare, and Business Analytics.

Why Do We Need Pandas?

Python provides basic data structures such as lists, tuples, and dictionaries. While these structures are useful for general programming, they are not efficient for working with large tabular datasets.

Pandas addresses these limitations by providing specialized data structures and built-in functions for data analysis. It allows users to perform complex operations with only a few lines of code.

Using Pandas, data scientists can:

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- Import datasets from multiple file formats
- Explore and understand datasets
- Clean missing or incorrect data
- Filter and sort records
- Combine multiple datasets
- Perform statistical analysis
- Summarize information
- Prepare data for visualization and machine learning

Data Structures in Pandas

Pandas provides two primary data structures:

- 1. Series:** A Series is a single column of data. Each element in a Series has an associated index.

Index	Value
0	85
1	90
2	78
3	95

Series are commonly used to represent a single column of data.

- 2. DataFrame:** A DataFrame is a two-dimensional table consisting of rows and columns. It is the most commonly used data structure in Pandas because it closely resembles a spreadsheet or database table.

Example:

Student Age Marks

Ali	20	85
Sara	21	90
Ahmed	22	78

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Each column can contain a different data type, making DataFrames highly flexible for real-world datasets.

Common Data Sources

Pandas can import and export data from various sources, including:

- CSV (Comma-Separated Values)
- Excel (.xlsx)
- JSON
- SQL Databases
- HTML Tables
- XML

This flexibility allows analysts to work with data collected from different applications and platforms.

Data Cleaning

Real-world datasets are rarely perfect. They often contain errors that can affect analysis and model performance.

Common data quality issues include:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid entries
- Inconsistent formatting

Pandas provides built-in functions to identify and correct these issues, improving the quality and reliability of the dataset.

Data Exploration

Before performing analysis, it is important to understand the dataset.

Pandas provides functions to:

- View the first and last records
- Check dataset dimensions
- Identify data types

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- Generate summary statistics
- Count missing values
- Examine unique values

This initial exploration helps analysts understand the structure and quality of the data.

Data Manipulation

Once data has been explored, it can be manipulated to meet analysis requirements.

Common operations include:

- Selecting specific rows and columns
- Filtering records based on conditions
- Sorting data
- Renaming columns
- Adding new columns
- Removing unnecessary columns
- Updating existing values

These operations enable analysts to transform raw data into a suitable format for further processing.

Applications of Pandas in Data Science

Pandas is widely used in various stages of the Data Science pipeline, including:

- Data collection and importing
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Statistical analysis
- Data visualization preparation
- Feature engineering
- Machine learning data preparation

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```
[2] ✓ On
#Check if NumPy and Pandas are Installed

import numpy as np
import pandas as pd

print("NumPy and Pandas are successfully imported!")

NumPy and Pandas are successfully imported!
```

If not installed, then install

```
• #Installing NumPy (if required)
!pip install numpy

*** Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

You can install both

```
• !pip install numpy pandas

*** Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

https://colab.research.google.com/drive/1R5GaQjCw9H-XWQea_vl9cLvWSW_eNca?usp=sharing

<https://colab.research.google.com/drive/1lbpvLqTwm1YdxMihKldw7Yj94xkTnoEp?usp=sharing>

access these notebooks for your practice and understanding

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Exercise

Q1. Create a DataFrame

Create a Pandas DataFrame using the following data:

Student Name Roll No Marks

Ali	101	85
Sara	102	90
Ahmed	103	78
Ayesha	104	92
Bilal	105	88

Perform:

- Display the DataFrame.
- Display the first 3 rows using head().
- Display the last 2 rows using tail().

```
[1] # Lab # 3: Introduction to Data Manipulation Using Pandas
# Student Name: Arsalan
# Roll No: 23BSCYS016

import pandas as pd
import numpy as np

[2] # Create the DataFrame
data_q1 = {
    'Student Name': ['Ali', 'Sara', 'Ahmed', 'Ayesha', 'Bilal'],
    'Roll No': [101, 102, 103, 104, 105],
    'Marks': [85, 90, 78, 92, 88]
}

df1 = pd.DataFrame(data_q1)

# 1. Display the DataFrame
print("\n--- Full DataFrame ---")
print(df1)

# 2. Display the first 3 rows
print("\n--- First 3 Rows (head) ---")
print(df1.head(3))

# 3. Display the last 2 rows
print("\n--- Last 2 Rows (tail) ---")
print(df1.tail(2))
```

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Output

```
... --- Full DataFrame ---
  Student Name  Roll No  Marks
0         Ali      101     85
1         Sara      102     90
2        Ahmed      103     78
3       Ayesha      104     92
4        Bilal      105     88

--- First 3 Rows (head) ---
  Student Name  Roll No  Marks
0         Ali      101     85
1         Sara      102     90
2        Ahmed      103     78

--- Last 2 Rows (tail) ---
  Student Name  Roll No  Marks
3       Ayesha      104     92
4        Bilal      105     88
```

Q2. DataFrame Information

Using the above DataFrame, find:

- Number of rows and columns using shape
- Column names
- Data types of columns

```
[1] # 1. Number of rows and columns
    shape_info = df1.shape
    print(f"Shape (Rows, Columns): {shape_info}")

    # 2. Column names
    print("\nColumn Names:")
    print(df1.columns.tolist())

    # 3. Data types of columns
    print("\nData Types:")
    print(df1.dtypes)

... Shape (Rows, Columns): (5, 3)

Column Names:
['Student Name', 'Roll No', 'Marks']

Data Types:
Student Name    object
Roll No         int64
Marks           int64
dtype: object
```

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Q3. Filtering Data

Using the student DataFrame:

1. Display only students who scored more than 85 marks.
2. Display the record of a student whose name is "Sara".

```
# 1. Display students who scored more than 85 marks
high_scorers = df1[df1['Marks'] > 85]
print("--- Students with Marks > 85 ---")
print(high_scorers)

# 2. Display the record of student named "Sara"
sara_record = df1[df1['Student Name'] == 'Sara']
print("\n--- Record for Sara ---")
print(sara_record)
```

```
*** --- Students with Marks > 85 ---
  Student Name  Roll No  Marks
1         Sara      102     90
3        Ayesha      104     92
4         Bilal      105     88

--- Record for Sara ---
  Student Name  Roll No  Marks
```

Q4. Handle Missing Values

Create a DataFrame:

Student Name Marks

Ali 85

Sara NaN

Ahmed 90

Ayesha NaN

Bilal 88

Perform:

- Check missing values.
- Replace missing marks with the **median value**.
- Display the updated DataFrame.

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```
Q. Create the DataFrame with missing values
data_dk = {
    "Student Name": ["Ali", "Sara", "Ahmed", "Ayesha", "Bilal"],
    "Marks": [85, np.nan, 88, np.nan, 88]
}

dfk = pd.DataFrame(data_dk)

# 1. Check missing values
print("\n--- Missing Values Count ---")
print(dfk.isnull().sum())

# 2. Impute missing marks with the median value
median_marks = dfk["Marks"].median()
dfk["Marks"].fillna(median_marks, inplace=True)

# 3. Display the updated DataFrame
print("\n--- Updated DataFrame (Missing values Imputed with Median) ---")
print(dfk)

--- Missing Values Count ---
Student Name    0
Marks           2
dtype: int64

--- Updated DataFrame (Missing Values Imputed with Median) ---
Student Name  Marks
0          Ali    85.0
1          Sara    88.0
2          Ahmed   88.0
3          Ayesha   88.0
4          Bilal    88.0

/home/ippuramul_3004/20220801.py:16: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 2.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.
For example, when doing "df[col].method(value, inplace=True)", try using "df.method(col, value, inplace=True)" or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

dfk["Marks"].fillna(median_marks, inplace=True)
```

Q5. Save and Read File

Using your DataFrame:

1. Save it as a CSV file.
2. Read the CSV file again using `pd.read_csv()`.
3. Display the data.

```
# 1. Save DataFrame as a CSV file
file_path = 'student_records_23bscys016.csv'
df1.to_csv(file_path, index=False)
print(f"Data saved to {file_path}")

# 2. Read the CSV file back into Pandas
df_loaded = pd.read_csv(file_path)

# 3. Display the loaded data
print("\n--- Data Read from CSV ---")
print(df_loaded)

*** Data saved to student_records_23bscys016.csv

--- Data Read from CSV ---
Student Name  Roll No  Marks
0          Ali    101    85
1          Sara    102    90
2          Ahmed   103    78
3          Ayesha   104    92
4          Bilal    105    88
```

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Colab link: https://colab.research.google.com/drive/13-75gh41tg_JDmG4WRz5sMS1IRsU4VS-?usp=sharing

Submission Checklist

Before submitting, make sure:

- ☐ Notebook name contains your **Name + Roll Number**
- ☐ All questions are solved
- ☐ Code output is visible
- ☐ Notebook is shared as **Viewer only**
- ☐ Link is submitted